

Data Converters

Exercise 06 – Analysis of a Switched Capacitor Filter and a Latch

Task I – Analysis of a Switched-Capacitor Filter

Shown in Figure 1 is a switched capacitor amplifier in the redistribution phase.

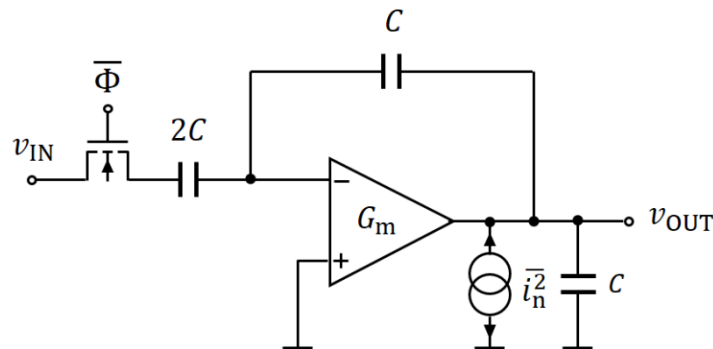


Figure 1: Switched capacitor amplifier.

The following parameters are given:

$$\begin{aligned} v_{IN} &= 0.5 \text{ V} & G_m &= 10 \text{ mS} & C &= 1 \text{ pF} \\ V_{th} &= 0.5 \text{ V} & C_{OX} &= 50 \text{ fF} & C_{OV} &= 10 \text{ fF} \\ \overline{i_n^2} &= n_f \cdot 4kT\gamma G_m & n_f &= 2 & \gamma &= 2/3 \end{aligned}$$

Note that the circuit is equivalent to the circuit analyzed in PE III and PE IV with $C_S = 2C$. The following findings can thus be reused:

$$\text{return factor } \beta = \frac{C_F}{C_F + C_S} \qquad \text{output pole } p = \frac{\beta G_m}{C_L + (1 - \beta)C_F}$$

Considering the output pole p , it can be seen that the output resistance of the amplifier due to feedback results in

$$r_{outF} = \frac{1}{\beta G_m}$$

and the total capacitance at the output node

$$C_{Ltot} = C_L + (1 - \beta)C_F$$

- Calculate the noise power due to the G_m -cell at the output of the amplifier, i.e., due to the output-referred current noise power spectral density $\overline{i_n^2}$ of the G_m -cell.
- Determine the on-resistance of the switch R_{on} so that it contributes no more than an additional $100 \mu V^2$ to the overall noise power at the output.

notes:

- $PSD_{R_{on}} = 4kTR_{on}$
 - The impact of R_{on} on the frequency response of the SC amplifier can be neglected.
- When the switch is turned off, i.e., Φ switches from high = 1.8 V to low = 0 V (fast gating), determine the disturbance of the output voltage v_{OUT} due to
 - channel charge injection
 - clock feedthrough

Task II – Analysis of a Latch

Let's analyze the small-signal behavior of the latch shown in Figure 2, which was published in *H.-S. Lee et al., "A self-calibrating 15 bit CMOS A/D converter," IEEE J. Solid-State Circuits, vol. 19, no. 6, pp. 813-819, June 1984.*

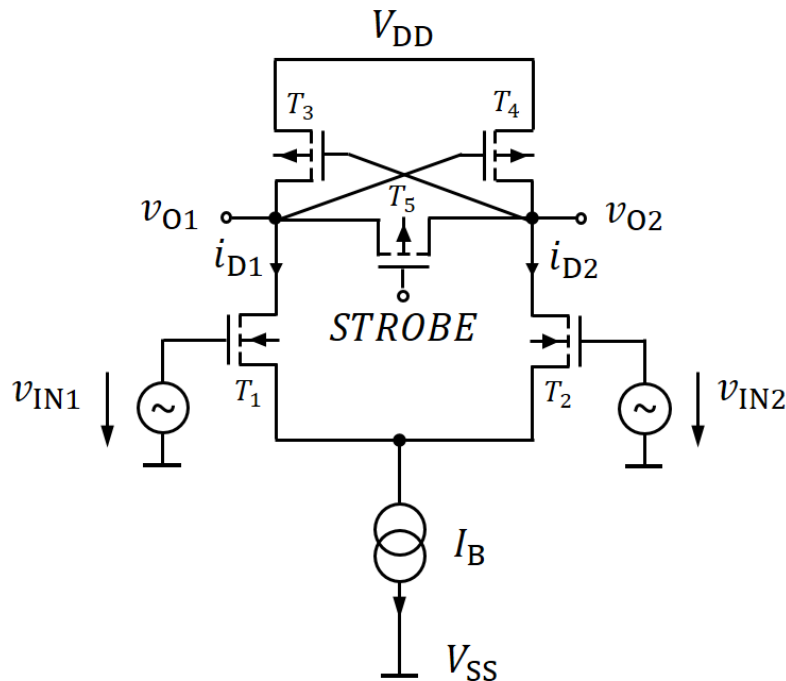


Figure 2: Latch.

In the comparison phase, i.e., T_5 is turned off since $STROBE = 1$, the circuit can be simplified with the model shown in Figure 3. The total capacitance at each output node is given by the load capacitor C_L .

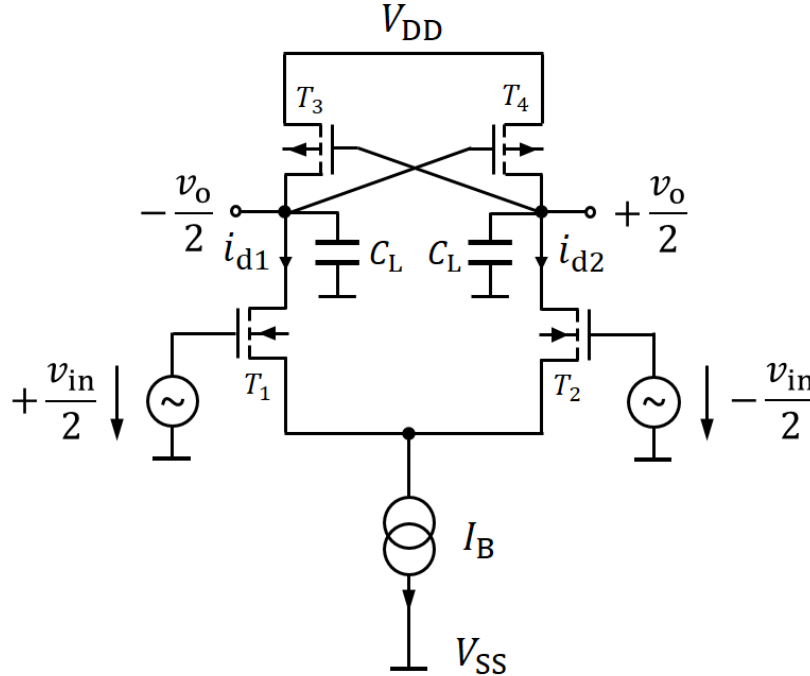


Figure 3: Small-signal model of the latch in the comparison phase.

For the small-signal analysis, the comparison phase is modelled by a step v_{in} at the time $t = 0$ s. All transistors are operating in the saturation region and T_1 and T_2 are always on.

- Draw the small-signal equivalent circuit of the left half circuit consisting of T_1 and T_3 .
- Derive an expression for the small-signal differential output voltage $v_o(t)$ in the time-domain for $t > 0$ s and $v_o(t = 0) = 0$ V.

Note: The antiderivative of the ordinary differential equation $\frac{dy(t)}{dt} + P \cdot y(t) = Q$ is $y(t) \cdot e^{\int P dt} = \int Q \cdot e^{\int P dt} dt + C$.

- The circuit follows the settling behavior derived in task II.b until either the p-MOS transistors T_3 or T_4 turns off. Determine the corresponding output voltage v_o as a function of the supply voltage V_{DD} , the output common-mode voltage $V_{CM,0}$, and the threshold voltage $V_{th,p}$ by considering the right-half circuit of the latch and T_3 .

Notes:

- $v_o = V_{O,CM} + \frac{v_o}{2}$

- Perfect symmetry can be assumed, i.e., $\frac{v_o}{2} = \left| -\frac{v_o}{2} \right|$
- d. Once either T_3 or T_4 turned off, slewing occurs. Determine the slew rate assuming that $i_{D1} = i_{D2} \approx I_B/2$, i.e., for very small differential input signals v_{in} .
 - e. Determine the overall time it takes the output $-v_o/2$ to transition over a voltage range Δv_{fall} starting at $V_{O,CM}$ based on your findings in the previous tasks.

Task III – DNL & INL of a Flash ADC

Consider a 7-bit Flash ADC with an ideal reference implemented as a resistor string and a full-scale range of 1 V. All comparators except for the first and the last have an offset voltage with a standard deviation $\sigma_{OS} = 2$ mV.

- a. Determine the DNL of each transition.
- b. Determine the INL of each transition.